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Iso-Damped Symmetric Fractional-Order Proportional-Integral-Derivative Control Using a Single Add-Differentiate Integrated Circuit Topology and Constant-Phase-Element Retuning

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Abstract
A fractional-order extension of a single Add-Differentiate Integrated Circuit (AD-IC) based proportional-integral-derivative (PID) controller is presented. The reference AD-IC topology realizes voltage-mode and current-mode integer-order PID controllers using a single active element, two resistors, and two capacitors. In this study, the voltage-mode realization is generalized by replacing the matched capacitive elements with constant phase element (CPE) operators of common order α, producing a symmetric fractional $\mathrm{PI}^{\alpha}\mathrm{D}^{\alpha}$ controller. The integer-order reference design is reconstructed, and direct fractional substitution at the original $Q = 80$ pF-equivalent scale is shown to destabilize all tested fractional cases. The instability arises because the original value is no longer compatible with the altered magnitude and phase characteristics introduced by the fractional-order CPE elements. A one-parameter retuning method is therefore introduced, where the CPE pseudo-capacitance scale $Q_{\mathrm{opt}}(\alpha)$ is selected to recover the reference phase margin (PM). With a ninth-order Oustaloup approximation, the retuned fractional controllers reduce settling time by up to 62.07% under equal phase margin. Under the stricter equal-phase-margin and equal-gain-crossover comparison, the settling-time improvement remains 27.37%, showing that the benefit is not solely caused by increased bandwidth. Robustness analysis further shows that the phase-margin standard deviation under plant-gain variation is reduced by 93.40% for $\alpha = 0.5$, while all simulated Monte Carlo cases remained closed-loop stable within the tested component, CPE, and plant uncertainty ranges. Finally, LTspice simulations agree closely with MATLAB for the integer-order and $\alpha = 0.5$ fractional designs, supporting the theoretical findings.

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1 Introduction

Proportional-integral-derivative (PID) control remains one of the most widely used feedback-control strategies because it combines structural simplicity, intuitive tuning, and effective error correction across many application domains [2, 19]. Classical tuning rules and modern PID surveys continue to emphasize the trade-off between response speed, robustness, and implementation simplicity [1, 49]. These properties have supported PID use across a wide range of practical control systems [26, 42].

Analog PID realizations have been developed using Operational Transconductance Amplifiers (OTAs) and Second-Generation Current Conveyors (CCIIs) [13, 47]. Voltage-mode designs based on Current Feedback Operational Amplifiers (CFOAs) and Voltage-Differencing Current Conveyors (VDCCs) provide further compact alternatives [3, 30]. Other compact structures employ versatile current-conveyor arrangements, electronically tunable log-domain current-mode circuits, and z-copy active elements [35, 38, 48]. More recent circuit realizations include single second-generation voltage conveyor (VCII) voltage-mode controllers, single voltage-differencing inverting buffered amplifier (VDIBA) current-mode controllers, and a mixed-mode CFOA implementation reported in 2024 [28, 31, 36]. These developments show the continuing emphasis on reducing active-element count while retaining useful tuning and operating-mode flexibility in analog PID circuits [6, 30].

A recent analog-circuit contribution introduced voltage-mode and current-mode PID controller realizations based on a single Add-Differentiate Integrated Circuit (AD-IC), two resistors, and two capacitors [24]. The broader AD-IC design family has also been used in compact analog building blocks such as mutative four-port and rectifier circuits [15, 16]. Related AD-IC-based circuit work has demonstrated its suitability for memory-element and memstance simulations [23]. The reference voltage-mode AD-IC PID study reported a gain crossover close to 2.37×10^7 rad/s and a phase margin (PM) of approximately 34.84° for a second-order low-pass plant [24]. Robust loop shaping has long been interpreted through gain, phase, and crossover behavior, which makes phase margin around crossover a natural design quantity for extending the AD-IC controller [2, 5].

Fractional-order control extends the integer-order PID idea by allowing the integral and derivative orders to be non-integer. The mathematical basis of fractional differentiation and integration predates its widespread use in modern control design [27]. The general $PI^{\lambda}D^{\mu}$ controller was established as a direct extension of classical PID control and subsequently developed in frequency-domain studies and monographs [25, 34, 41]. Fractional-order control monographs further consolidated the modeling, simulation, and robust-control framework for these systems [33, 37]. Tuning rules based on response data, crossover conditions, and robustness specifications have been reported in several forms [9, 21, 40]. Optimization-based rules extend this line of work to difficult process classes [32], while more recent studies have introduced data-driven

fractional PID design procedures [46]. Recent application work continues to exploit phase-margin, crossover, and iso-damping objectives in fractional PID design [45], while a 2025 survey documents the continuing expansion of fractional PID architectures, tuning strategies, and applications [4].

In analog-circuit settings, fractional-order elements are commonly represented using constant phase elements (CPEs), fractance networks, or finite-band rational approximations [14, 43]. Fractional-order circuits and systems have been reviewed as an emerging circuit-design area, with CPE emulation and passive fractance realization forming important implementation routes [12, 17]. Recent Simulation Program with Integrated Circuit Emphasis (SPICE) work has also demonstrated efficient Foster-type resistor-capacitor (RC) representations of fractional capacitors for finite-band time-domain simulation [44]. Recursive rational approximations, including Oustaloup-type methods, are widely used to simulate fractional operators over finite frequency bands [22, 29]. Approximation-based fractional control must therefore be interpreted in relation to the selected frequency band and approximation order [22]. Fractal-system representations provide another classical view of fractional dynamics that motivates finite-order approximations in circuit and control simulations [8].

This paper addresses whether the compact single-AD-IC voltage-mode PID topology can be extended to a fractional-order controller by replacing its matched capacitive branches with CPE-like operators. The novelty differs from conventional fractional PID formulations and tuning studies, where the controller gains and fractional orders constitute multiple independent design variables [4, 34]. Here, the circuit topology itself imposes a symmetric order and preserves the resistor-set proportional term, leaving the matched CPE scale as the only continuous retuning variable for a chosen α. Direct substitution is shown to be insufficient and can destabilize the loop, so a topology-specific CPE pseudo-capacitance retuning method is introduced to recover the integer-design phase margin without increasing the number of active elements. To separate phase-shaping benefits from the trivial effect of increased bandwidth, an equal-phase-margin and equal-gain-crossover comparison is also included. The resulting contribution is therefore not a new generic fractional PID tuning rule, but a topology-preserving route for adding fractional phase shaping and iso-damping to the single-AD-IC architecture.

The main contributions are as follows:

- A symmetric $\mathrm{PI}^{\alpha}\mathrm{D}^{\alpha}$ transfer function is derived for the voltage-mode AD-IC topology, and the invariance of $K_p = 1 + R_1/R_2$ under matched CPE substitution is identified.
- Direct CPE substitution at fixed Q is shown to destabilize all tested fractional cases, establishing that explicit retuning is required.
- A one-parameter retuning method based on $Q_{\mathrm{opt}}(\alpha)$ is introduced, and a dense log-linear design map is obtained.
- Nominal, equal-crossover, plant-gain, approximation-order, and Monte Carlo robustness tests are used to separate transient improvement, phase-shaping, and approximation effects.

- Behavioral LTspice validation is added using ideal AD-IC port equations and finite RC CPE emulation, providing circuit-netlist evidence consistent with the MATLAB model.
- A transistor-level PSpice frequency-response check is performed using the Taiwan Semiconductor Manufacturing Company (TSMC) 0.25 μm Berkeley Short-channel Insulated-Gate Field-Effect Transistor Model, version 3 (BSIM3) AD-IC subcircuit, providing circuit-level evidence that the characteristic PID gain and phase trends are retained for both the integer and fractional designs.

Table 1 positions the proposed controller relative to representative recent and established analog PID realizations and to a recent fractional PID study that explicitly considers iso-damping. The comparison separates circuit compactness from the control-level robustness objective: several analog PID circuits achieve low active-element count or convenient tunability, whereas iso-damping is generally not evaluated at circuit level.

2 Reference AD-IC PID Controller and Fractional Extension

2.1 Integer AD-IC PID Realization

The reference voltage-mode AD-IC PID realization considered here corresponds to the voltage-mode realization reported in the original AD-IC PID study [24]. Its transfer function can be written as

$$\frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{R_1R_2C_1C_2s^2 + (R_2C_2 + R_1C_1)s + 1}{sC_1R_2}. \tag{1}$$

Separating the derivative, proportional, and integral terms gives

$$G_c(s) = R_1C_2s + \left(\frac{C_2}{C_1} + \frac{R_1}{R_2}\right) + \frac{1}{sC_1R_2}. \tag{2}$$

For the matched-capacitor case $C_1 = C_2 = Q$, Eq. (2) reduces to

$$G_c(s) = R_1Qs + \left(1 + \frac{R_1}{R_2}\right) + \frac{1}{R_2Qs}. \tag{3}$$

Thus,

$$K_d = R_1Q, \qquad K_p = 1 + \frac{R_1}{R_2}, \qquad K_i = \frac{1}{R_2Q}. \tag{4}$$

The component values used in the reference design are $R_1 = 500\,\Omega$, $R_2 = 3000\,\Omega$, and $Q = 80$ pF, which give $K_d = 4.0000 \times 10^{-8}$, $K_p = 1.1667$, and $K_i = 4.1667 \times 10^6$.

Table 1 Comparison with representative PID and fractional-PID literature

Reference	Controller / mode	Active element	Active count	Passive elements	Tunability characteristic	Iso-damping
Ayten et al. [3]	Integer PID / voltage mode	CFOA	1	2R+2C	Passive R/C adjustment	NR
Özer and Kaçar [30]	Integer PID / voltage mode	VDCC	1	4R+2C	Electronic control parameters plus R/C design	NR
Özer et al. [31]	Integer PID / voltage mode	VCII	1	2R+2C	Passive R/C adjustment	NR
Oruçoğlu et al. [28]	Integer PID / current mode	VDIBA	1	3R+2C	Controller parameters adjustable without topology change	NR
Roongmuanpha et al. [36]	Integer PID / mixed mode	CFOA	3	4R+2C	K_P, K_I, and K_D adjustable by component values	NR
Minayi and Göknar [24]	Integer PID / voltage and current modes	AD-IC	1	2R+2C	Passive R/C adjustment	NR
Ynineb et al. [45]	Fractional PID / model-based	N/A	N/A	N/A	Controller tuning to phase-margin, cutoff-frequency, and robustness specifications	Yes
This work	Fractional $PI^{\alpha}D^{\alpha}$ / voltage mode	AD-IC	1	2R+2CPE	One-parameter $Q_{\mathrm{opt}}(\alpha)$ retuning for chosen α	Yes

Passive-element counts refer to the reported controller core. For the present work, "2R+2CPE" is the topology-level count; each CPE is replaced by a finite RC network in a physical rational realization, so the final resistor/capacitor count depends on approximation order. NR denotes "not reported," and N/A denotes "not applicable"

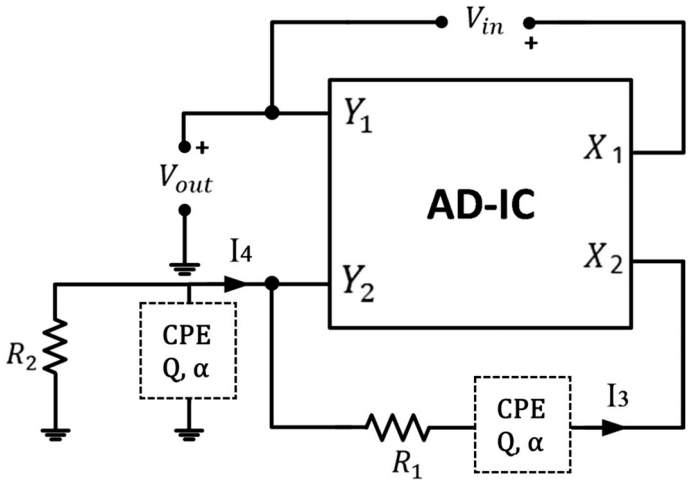


Fig. 1 Fractional-order extension of the voltage-mode AD-IC PID realization. The two matched capacitive branches of the integer topology are replaced by matched CPE blocks with $Y_{\mathrm{CPE}}(s) = Qs^{\alpha}$. The matched choice $Q_1 = Q_2 = Q$ and $\alpha_1 = \alpha_2 = \alpha$ preserves the resistor-set proportional gain $K_p = 1 + R_1/R_2$

2.2 Matched-CPE Fractional AD-IC Topology

The fractional topology used in this study is shown in Fig. 1. The two matched capacitors in the integer AD-IC realization are replaced by matched CPE blocks with common pseudo-capacitance scale Q and common order α. A CPE is represented in the Laplace domain by the admittance

$$Y_{\mathrm{CPE}}(s) = Qs^{\alpha}, \qquad 0 < \alpha \leq 1. \tag{5}$$

For $\alpha = 1$, Q becomes an ordinary capacitance. For $\alpha < 1$, Q is not dimensionally identical to a standard capacitance and is interpreted as a CPE pseudo-capacitance scale.

Under the matched CPE replacement, Eq. (3) becomes

$$G_{c,\alpha}(s) = R_1 Qs^{\alpha} + \left(1 + \frac{R_1}{R_2}\right) + \frac{1}{R_2 Qs^{\alpha}}. \tag{6}$$

Equivalently,

$$G_{c,\alpha}(s) = K_d s^{\alpha} + K_p + K_i s^{-\alpha}, \tag{7}$$

where

$$K_d = R_1 Q, \qquad K_p = 1 + \frac{R_1}{R_2}, \qquad K_i = \frac{1}{R_2 Q}. \tag{8}$$

The proportional-gain invariance follows from the ratio of the two matched fractional admittances. For $Q_1 = Q_2 = Q$ and $\alpha_1 = \alpha_2 = \alpha$,

$$\frac{Q_2 s^{\alpha}}{Q_1 s^{\alpha}} + \frac{R_1}{R_2} = 1 + \frac{R_1}{R_2}. \tag{9}$$

Thus, the proportional contribution remains the same resistor-set quantity as in the integer circuit, while the derivative-like and integral-like terms become $R_1 Q s^{\alpha}$ and $(R_2 Q s^{\alpha})^{-1}$, respectively. If either the CPE scales or the CPE orders were not matched, the admittance ratio would become

$$\frac{Q_2 s^{\alpha_2}}{Q_1 s^{\alpha_1}} = \frac{Q_2}{Q_1} s^{\alpha_2 - \alpha_1}, \tag{10}$$

which introduces an additional frequency-dependent term into the proportional path. The symmetric $\lambda = \mu = \alpha$ form therefore follows from the matched $C_1 = C_2$ reference circuit and preserves the single-parameter retuning structure that is central to the proposed AD-IC extension. A nonsymmetric $PI^{\lambda}D^{\mu}$ implementation may be useful in future work, but it would require two different fractional elements in the two branches. That would increase implementation complexity, tolerance sensitivity, and the number of retuning variables. The present design deliberately preserves the matched-element philosophy of the reference AD-IC circuit and asks how far that structure can be extended before additional fractional degrees of freedom are needed.

2.3 Finite RC Approximation of the CPE

Physical CPEs are normally implemented only over a finite frequency band. Figure 2 illustrates the RC ladder interpretation used to connect the ideal CPE model to circuit-level implementation. Each additional RC section increases the approximation degree, so the order N controls the number of sections used to emulate the target fractional impedance or admittance over the selected frequency interval. Passive RC fractance networks and constant-argument immittance methods provide classical approximations of fractional behavior [7, 10]. Active CPE emulators and biquadratic approximation approaches provide alternative routes when compactness or tunability is required [11, 39].

The finite realization is not a secondary implementation detail because it defines what is meant by fractional order in a circuit simulator. An ideal CPE has constant phase over all frequencies, whereas any RC ladder can only approximate that behavior over a selected interval. Therefore, the approximation band must cover the controller crossover region and the dominant transient frequencies of the closed-loop response. In this study, the band $[10^3, 10^9]$ rad/s was selected to include the nominal gain-crossover frequencies, the local phase-shaping region, and the visible transient oscillation frequencies. The approximation-order checks in Fig. 9 and the local phase-error check in Fig. 10 are included specifically to make this finite-band assumption explicit.

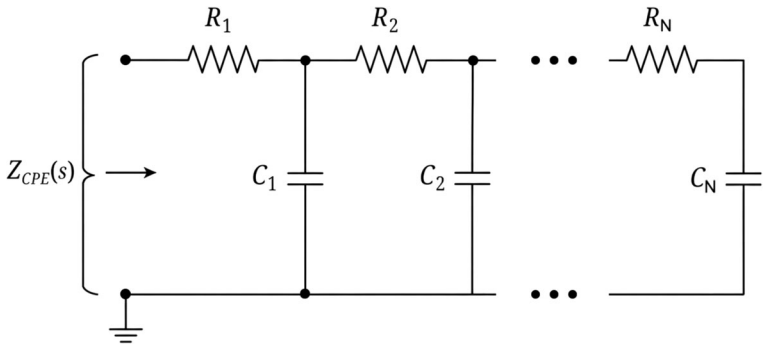


Fig. 2 Finite RC approximation concept for a CPE. The approximation degree corresponds to the number of RC sections used to emulate the target $Z_{\mathrm{CPE}}(s)$ or $Y_{\mathrm{CPE}}(s)$ over a finite frequency range. In the simulations, the finite approximation was generated from the same ninth-order Oustaloup rational model used in the MATLAB controller analysis

For circuit interpretation, the rational CPE admittance can be expanded into a passive parallel Foster form,

$$Y_N(s) \approx G_0 + \sum_{k=1}^{N_b} \frac{sC_k}{1 + sR_kC_k}, \tag{11}$$

where each term represents one series RC branch and G_0 is an optional leakage conductance. The branch parameters are obtained from the residues and poles of the rational approximation. If a pole is located at $-p_k$ and the associated residue contribution is $r_k/(s + p_k)$, the corresponding branch conductance can be written as $g_k = -r_k/p_k$, with

$$R_k = \frac{1}{g_k}, \qquad C_k = \frac{g_k}{p_k}. \tag{12}$$

This conversion gives a direct route from the MATLAB fractional operator to an LTspice-compatible RC subcircuit. It also clarifies the meaning of "degree" in Fig. 2: increasing the approximation degree increases the number of pole-zero or RC sections used to reproduce the CPE response over the design band.

2.4 Controlled Plant and Closed-Loop Definitions

The second-order low-pass plant used in the original AD-IC PID paper was adopted:

$$G_p(s) = \frac{80 \times 10^{14}}{20s^2 + 10^7 s + 10^{14}}. \tag{13}$$

The open-loop and unity-feedback closed-loop transfer functions are

$$L_\alpha(s) = G_{c,\alpha}(s)G_p(s), \qquad T_\alpha(s) = \frac{L_\alpha(s)}{1 + L_\alpha(s)}. \tag{14}$$

The integer baseline was first reconstructed to ensure that the plant, feedback sign, and PID coefficients reproduced the reference controlled-plant behavior before any fractional-order modification was introduced.

2.5 Fractional Approximation and Design Cases

Closed-loop time-domain and frequency-domain calculations were performed using rational Oustaloup-type approximations of s^{α} over $[10^3, 10^9]$ rad/s [22, 29]. The main manuscript results used approximation order $N = 9$, while sensitivity checks used $N = 5, 7, 9, 11, 13$. The approximation can be written as

$$s^{\alpha} \approx K_N \prod_{k=1}^{N} \frac{s + \omega_{z,k}}{s + \omega_{p,k}}, \tag{15}$$

where K_N is selected to match the exact magnitude at the geometric center of the approximation band. The default $N = 9$ value was selected as a middle-order engineering compromise within the explicit $N = 5, 7, 9, 11, 13$ sensitivity sweep: it is sufficiently above the lower-order cases to avoid basing the conclusions on a coarse approximation, while keeping the rational transfer-function order and the corresponding RC-emulator complexity below the $N = 11$ and $N = 13$ alternatives. Figure 9 verifies that the main ordering of the controller responses is unchanged across the tested approximation orders, so the reported performance trend is not contingent on the single choice $N = 9$.

A finite-band Oustaloup model also differs from an ideal fractional operator at direct current (DC). For Eq. (15), its DC gain is

$$G_N(0) = K_N \prod_{k=1}^{N} \frac{\omega_{z,k}}{\omega_{p,k}}, \tag{16}$$

which is finite and nonzero because all poles and zeros of the rational approximation are finite. Consequently, the reciprocal approximation used for $s^{-\alpha}$ also has a finite DC gain, $G_N^{-1}(0) = 1/G_N(0)$, instead of the infinite DC gain of an ideal fractional integrator. This is the finite-DC-gain effect referred to later in the time-domain robustness discussion; it is a property of truncating the fractional operator to a rational model over a finite frequency interval, not of the ideal CPE itself.

Four orders were evaluated: $\alpha \in \{1.0, 0.9, 0.7, 0.5\}$. Three design cases were considered. First, direct substitution was tested using the original $Q = 80$ pF scale for all orders. Second, an equal-phase-margin design was introduced. For each α, Q was selected so that the phase margin (PM) of $L_{\alpha}(s)$ matched the integer reference value $\mathrm{PM}_{\mathrm{target}} = 34.78°$. Third, an equal-PM and equal-gain-crossover design was used as a fairness test. In that case, Q and a scalar loop-gain multiplier K_{loop} were selected so that

$$\angle L_{\alpha}(j\omega_{\mathrm{cp}}^{\star}) = -180° + \mathrm{PM}_{\mathrm{target}}, \qquad |K_{\mathrm{loop}} L_{\alpha}(j\omega_{\mathrm{cp}}^{\star})| = 1, \tag{17}$$

where $\omega_{cp}^{\star} = 2.3776 \times 10^7$ rad/s is the integer baseline gain-crossover frequency. The scalar K_{loop} is applied multiplicatively after selecting Q and only enforces the magnitude condition at $\omega_{cp}^{\star}$; it does not change the phase at that frequency.

2.6 Retuning as a One-Dimensional Design Problem

The matched-CPE structure reduces the fractional redesign to a one-dimensional search in the pseudo-capacitance scale Q for each selected order α. This is useful from a circuit-design perspective because R_1 and R_2 continue to set the proportional contribution, whereas Q controls the relative balance of the fractional derivative-like and integral-like branches. For a fixed α, the retuning objective can be written as

$$F_{\alpha}(Q) = \mathrm{PM}\left\{G_{c,\alpha}(s;\, Q)G_p(s)\right\} - \mathrm{PM}_{\mathrm{target}}. \tag{18}$$

The equal-PM design is obtained by solving $F_{\alpha}(Q) = 0$ over a physically meaningful interval of positive Q. The direct-substitution result in Table 3 shows why this search is necessary: the original 80 pF scale does not preserve the loop phase when the order is changed. Retuning Q shifts the controller zero-pole balance in the rational approximation and recovers the desired phase at the new gain crossover.

The design workflow used throughout the paper is therefore:

1. Reconstruct the integer AD-IC PID baseline from the reported component values and plant model.
2. Select a fractional order α and construct the finite-band rational approximation of s^{α}.
3. Solve Eq. (18) for $Q_{\mathrm{opt}}(\alpha)$ so that the reference PM is recovered.
4. Simulate the equal-PM closed-loop response and record transient metrics.
5. For the fairness case, adjust Q and K_{loop} so that both PM and the integer gain-crossover frequency are matched.
6. Validate robustness under plant-gain variation, Monte Carlo perturbations, and approximation-order variation.
7. Reproduce the selected integer and fractional cases in LTspice using ideal AD-IC port equations and finite RC CPE emulation.
8. Perform a transistor-level open-loop frequency-response check in PSpice using the BSIM3 AD-IC subcircuit to confirm circuit-level consistency.

This sequence separates three effects that would otherwise be conflated: the instability caused by naive substitution, the nominal acceleration caused by equal-PM retuning, and the residual fractional benefit that remains after crossover matching.

2.7 Robustness and LTspice Validation

Robustness was assessed using deterministic plant-gain sweeps, Monte Carlo uncertainty analysis, approximation-order sensitivity, and crossover-region approximation-error checks. The plant gain was swept from -18 to $+18$ decibels (dB), and PM was recorded for each controller. The Monte Carlo analysis used 300 independent

uncertainty trials for each reported α value. Trials used $\pm 5\%$ perturbations in R_1 and R_2, $\pm 10\%$ perturbations in Q, ± 0.02 perturbations in α for fractional cases, ± 12 dB plant-gain variation, and $\pm 10\%$ perturbations of each plant denominator coefficient.

A circuit-netlist validation was also performed in LTspice. The AD-IC was implemented with ideal behavioral voltage sources satisfying

$$V_{x1} = V_{y1} - V_{y2}, \qquad V_{x2} = V_{y1} + V_{y2}. \tag{19}$$

These equations preserve the ideal AD-IC port algebra used in the analytical derivation while avoiding the need for a transistor-level AD-IC model. For $\alpha = 1.0$, the two CPE branches reduce to ordinary 80 pF capacitors. For $\alpha = 0.5$, each CPE was implemented using a finite RC realization derived from the same $N = 9$ Oustaloup rational approximation used in MATLAB.

The plant was implemented in LTspice with a state-space behavioral realization of Eq. (13). The time-domain plant equation is

$$20\ddot{y} + 10^7\dot{y} + 10^{14}y = 80 \times 10^{14}u, \tag{20}$$

or, after normalization,

$$\ddot{y} + 5 \times 10^5\dot{y} + 5 \times 10^{12}y = 4 \times 10^{14}u. \tag{21}$$

A scaled velocity state $v_s = \dot{y}/10^7$ was used for numerical conditioning, giving

$$\dot{y} = 10^7 v_s, \qquad \dot{v}_s = 4 \times 10^7 u - 5 \times 10^5 v_s - 5 \times 10^5 y. \tag{22}$$

These equations were implemented using unit capacitors and behavioral current sources. The validation is therefore described as behavioral circuit validation, not as transistor-level silicon validation. It is nevertheless comparable in spirit to the SPICE-level check reported for the original integer AD-IC PID controller [24]. A complementary transistor-level frequency-response validation using the BSIM3 AD-IC subcircuit in PSpice is presented in Sect. 3.8.

3 Results

3.1 Integer Reconstruction and Direct Substitution

The reconstructed integer-order controller closely reproduced the reference AD-IC PID controlled-plant response. Table 2 shows that the maximum error was below 0.26%, confirming consistency with the original plant and controller configuration.

Table 3 shows that direct substitution at fixed $Q = 80$ pF is not a valid design method. The integer case remains stable, but all tested fractional cases produce negative PM values. Gain margin (GM) values are reported where meaningful.

Table 2 Integer AD-IC PID baseline validation against the reference controlled-plant response

Metric	This study	Reference	Absolute error	Percent error
Rise time [ns]	45.405	45.520	0.115	0.254
Peak time [ns]	120.410	120.700	0.288	0.239
Overshoot [%]	45.332	45.290	0.042	0.092
Settling time [μs]	0.64446	0.64600	0.00154	0.238

Table 3 Direct substitution test with fixed $Q = 80$ pF-equivalent scale

α	Stable	PM [deg]	GM [dB]	ω_{cp} [rad/s]	ω_{cg} [rad/s]
1.0	Yes	34.84	−14.92	2.378×10^7	9.419×10^6
0.9	No	−26.84	16.70	2.493×10^7	6.055×10^7
0.7	No	−55.53	−92.70	7.381×10^7	2.371×10^6
0.5	No	−36.59	−115.48	3.113×10^8	2.502×10^6

Fractional step-response metrics are not reported because the closed-loop systems are unstable

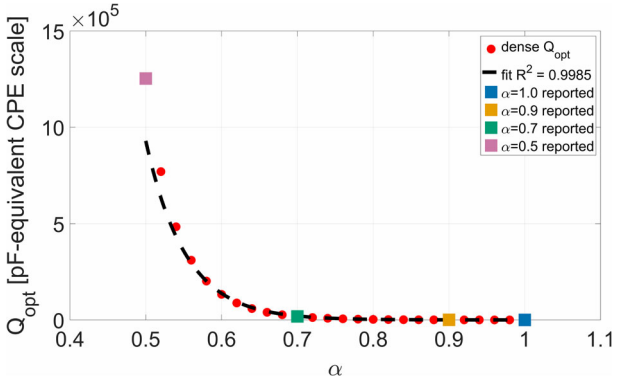


Fig. 3 Dense equal-PM CPE design map. The optimized Q_{opt} values follow a strong log-linear trend as a function of $1-\alpha$. The square markers show the four reported controller orders used in the main comparisons

3.2 CPE Retuning Law and Nominal Performance

The equal-PM retuning method selects $Q_{opt}(\alpha)$ so that each fractional controller recovers the reference PM. Figure 3 shows the dense design map over $\alpha = 0.50, 0.52, \ldots, 1.00$. The fitted relationship is

$$\log_{10}\left(Q_{opt}[\text{pF-equivalent}]\right) = 8.2738(1-\alpha) + 1.8312, \tag{23}$$

with $R^2 = 0.9985$. Here, R^2 is the coefficient of determination of the least-squares fit in the transformed coordinates $\log_{10}(Q_{opt})$ and $1-\alpha$; a value close to unity indicates

Table 4 Equal-PM retuning values and nominal closed-loop performance

α	Q_{opt} [pF-eq.]	Rise [ns]	Peak [ns]	Overshoot [%]	Settling [μs]	ω_{cp} [rad/s]
1.0	80.000	45.405	120.410	45.332	0.64446	2.3776×10^7
0.9	464.423	41.404	111.010	45.325	0.56446	2.5929×10^7
0.7	18300.964	35.404	94.209	44.438	0.36044	3.0908×10^7
0.5	1252931.862	24.402	64.606	43.186	0.24442	4.5381×10^7

The unit label pF-eq. denotes pF-equivalent

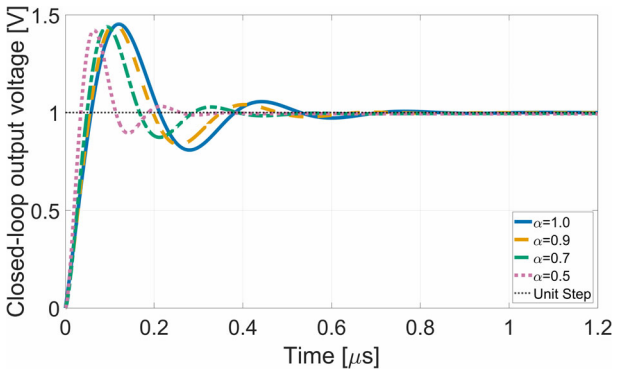


Fig. 4 Closed-loop step responses under equal PM. Lower fractional orders reduce the settling time after $Q_{\text{opt}}(\alpha)$ retuning while maintaining the target PM. The output is shown as voltage because the reference step is a 1 V input

that the fitted line explains nearly all of the variation of the computed design-map points in those coordinates. The values used for the reported orders are listed in Table 4.

The corresponding closed-loop step responses are shown in Fig. 4. Relative to the integer baseline, the settling-time reductions are 12.41%, 44.07%, and 62.07% for $\alpha = 0.9$, $\alpha = 0.7$, and $\alpha = 0.5$, respectively. This confirms that equal-PM fractional retuning can accelerate the nominal response while keeping the reference stability margin.

3.3 Equal-Crossover Fairness Comparison

The equal-PM comparison increases the gain-crossover frequency as α decreases, so a stricter equal-PM and equal-ω_{cp} comparison was performed. Figure 5 and Table 5 show the resulting responses. The scalar K_{loop} forces the common crossover condition after Q sets the phase condition.

The equal-crossover comparison reduces the apparent speed advantage, as expected. The rise times of the $\alpha = 0.7$ and $\alpha = 0.5$ designs are slightly longer than the integer baseline, increasing from 45.405 ns to 46.205 ns and 46.605 ns, respectively. Therefore, the remaining benefit should not be interpreted as improved initial rise. Instead, the fractional controllers improve settling by reshaping the post-peak response and reducing the later oscillatory tail. The $\alpha = 0.7$ and $\alpha = 0.5$ cases still reduce

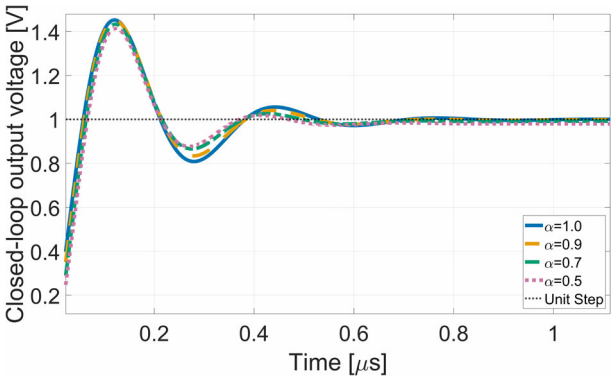


Fig. 5 Closed-loop step responses under the equal-PM and equal-gain-crossover fairness condition. The fractional designs retain a settling-time benefit even when the nominal crossover frequency is matched to the integer baseline

Table 5 Equal-PM and equal-gain-crossover closed-loop performance

α	Q [pF-eq.]	K_{loop}	Rise [ns]	Overshoot [%]	Settling [μs]
1.0	80.000	1.0000	45.405	45.332	0.64446
0.9	499.706	0.8427	45.405	45.131	0.61906
0.7	21479.788	0.6000	46.205	44.248	0.46805
0.5	1485446.120	0.3085	46.605	44.539	0.47125

The target gain-crossover frequency was $\omega_{\text{cp}}^{\star} = 2.3776 \times 10^7$ rad/s for all rows

settling time by approximately 27%, indicating that the improvement is not purely a bandwidth artifact.

3.4 Frequency-Domain Mechanism and Iso-Damping

The open-loop Bode response around crossover is shown in Fig. 6. This focused view is more informative than a full-band plot because the equal-PM condition and the phase-shaping mechanism are defined at the 0 dB crossing. The crossover markers lie near the target phase level $-180^\circ + \text{PM}_{\text{target}} = -145.22^\circ$, confirming that the retuned designs recover the intended PM. The lower-order fractional controllers also show a flatter local phase profile around crossover, which is consistent with iso-damping behavior.

The gain-sweep robustness in this study can be interpreted through the local open-loop phase slope,

$$S_\phi = \left. \frac{d\angle L(j\omega)}{d \log_{10} \omega} \right|_{\omega=\omega_{\text{cp}}}, \tag{24}$$

whose units are degrees per decade. This quantity is related to, but is not itself, a conventional dimensionless relative sensitivity. The deterministic gain sweep is a finite-perturbation robustness test: multiplying the plant gain shifts the frequency at

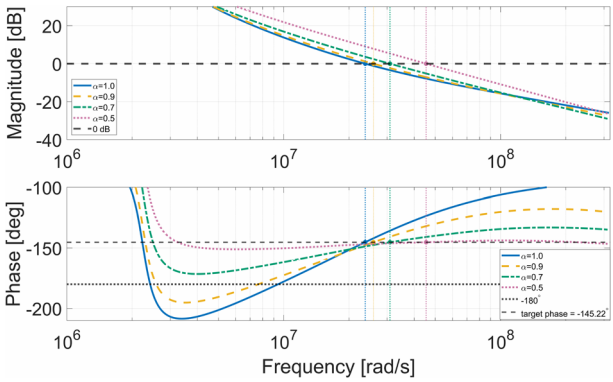


Fig. 6 Equal-PM open-loop Bode response around the gain-crossover region. The 0 dB line, gain-crossover markers, $-180°$ reference, and target phase level are shown to verify PM recovery and local phase reshaping

which the open-loop magnitude crosses 0 dB, and the resulting PM is recomputed at that shifted crossover. Locally,

$$\Delta \mathrm{PM} \approx S_\phi \, \Delta \log_{10} \omega_{\mathrm{cp}}, \tag{25}$$

so a small magnitude of S_ϕ means that a crossover displacement produces only a small PM displacement. This is the classical Bode loop-shaping mechanism underlying the iso-damping interpretation [5, 20]: if the phase changes rapidly around crossover, a plant-gain change produces a large PM change, whereas a locally flat phase curve keeps the shifted crossover close to the nominal phase. The gain-sweep curves therefore quantify the finite change directly, while Eq. (24) provides its local differential explanation.

Figure 7 shows PM under plant-gain variation from -18 to $+18$ dB. The integer baseline exhibits a strong PM change and becomes unstable for sufficiently reduced gain. By contrast, the $\alpha = 0.5$ curve remains nearly flat around the target PM.

The quantitative robustness summary is provided in Table 6. The PM standard deviation decreases from 26.644° for $\alpha = 1.0$ to 1.759° for $\alpha = 0.5$, corresponding to a 93.40% reduction. Monte Carlo trials also produced 100% closed-loop stability for all tested designs.

3.5 Time-Domain Response Under Plant-Gain Variation

To complement the frequency-domain PM sweep in Fig. 7, the closed-loop step response was evaluated at three representative gain conditions: $K = 0.25\times$ (-12 dB), $K = 1.0\times$ (nominal), and $K = 4.0\times$ ($+12$ dB). Both controllers were applied without retuning from their nominal equal-PM designs. Table 7 lists the time-domain and frequency-domain metrics for all six controller–plant combinations.

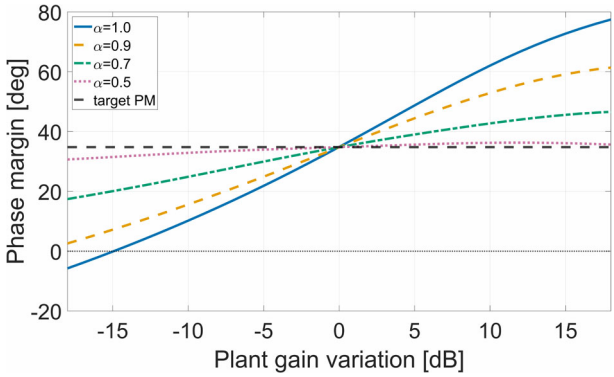


Fig. 7 Iso-damping robustness under plant-gain variation. Lower fractional orders substantially reduce PM sensitivity to plant-gain changes. The $\alpha = 0.5$ design maintains an almost flat PM profile over the tested gain range

Table 6 Robustness summary under deterministic plant-gain variation and Monte Carlo uncertainty

α	σ_{PM} [deg]	PM range [deg]	GS stable [%]	MC PM [5th,95th] [deg]	MC stable [%]
1.0	26.644	83.218	89.189	[7.170, 64.057]	100.0
0.9	19.050	58.803	100.000	[8.742, 60.775]	100.0
0.7	9.304	29.173	100.000	[19.741, 46.438]	100.0
0.5	1.759	5.628	100.000	[28.817, 38.686]	100.0

The quantity σ_{PM} denotes the phase-margin standard deviation, GS denotes the deterministic gain-sweep stability rate, and MC denotes Monte Carlo. Each MC row summarizes 300 independent uncertainty trials for the corresponding α value

Table 7 Closed-loop step response metrics under plant-gain variation ($K \times G_p$)

K	α	PM [°]	Rise [ns]	Peak [ns]	Overshoot [%]	Settling [μs]
0.25×	1.0	5.7	93	258	76.4	7.80
	0.5	37.4	52	133	39.1	22.4†
1.0× (nominal)	1.0	34.8	45	120	45.3	0.644
	0.5	34.8	22	57	41.1	0.206
4.0×	1.0	66.8	19	52	21.3	0.117
	0.5	37.2	9	24	38.9	0.093

Both controllers are designed for the nominal plant ($K = 1.0\times$, PM = 34.78°) and applied without retuning. † A small steady-state error is present because the finite Oustaloup direct-current (DC) gain defined in Eq. (16) is not infinite for the reciprocal fractional-integrator approximation, unlike an ideal fractional integrator

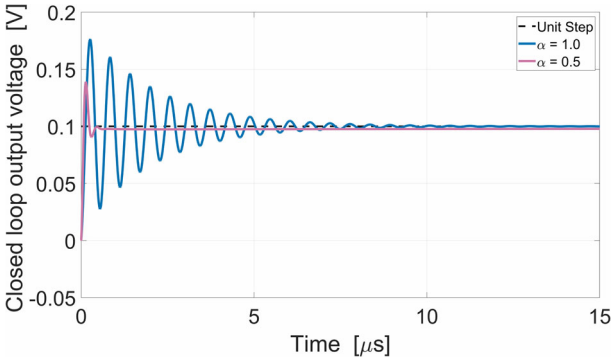


Fig. 8 Closed-loop step response under a fourfold plant gain reduction ($K = 0.25\times$, -12 dB). Both controllers are applied without retuning from their nominal designs. The integer $\alpha = 1.0$ controller operates near the stability boundary (PM = 5.7°) and produces sustained oscillations that take approximately 7.8 μs to decay. The fractional $\alpha = 0.5$ controller maintains PM = 37.4° and a bounded, decaying response, demonstrating the iso-damping robustness property in the time domain. A small residual steady-state error is present in the $\alpha = 0.5$ curve due to the finite DC gain of the Oustaloup rational approximation

Figure 8 shows the step responses for the most critical case, $K = 0.25\times$. With the integer controller, the loop operates at PM = 5.7°, producing sustained oscillations that persist for approximately 7.8 μs. The fractional $\alpha = 0.5$ controller maintains PM = 37.4° and produces a bounded, monotonically decaying response despite the same fourfold plant gain reduction. The overshoot of the fractional controller (39.1%) is close to its nominal value (41.1% at $K = 1.0\times$), consistent with the iso-damping property in the time domain.

The overshoot of the $\alpha = 0.5$ controller in Table 7 is consistently near 39–41% across all three gain conditions, while the integer design ranges from 21.3% to 76.4%. This time-domain consistency reflects the same frequency-domain phenomenon identified in Fig. 7: because the open-loop phase at the gain crossover changes little despite the shift in crossover frequency, the effective closed-loop damping and overshoot also remain nearly constant. The overall PM variation across the three tested conditions is 61.1° for $\alpha = 1.0$ and 2.6° for $\alpha = 0.5$.

The extended settling time of the $\alpha = 0.5$ design at $K = 0.25\times$ (22.4 μs) is attributable to the finite DC gain of the Oustaloup rational approximation rather than to marginal closed-loop stability. As defined in Eq. (16), the finite-order rational model has finite pole and zero frequencies and therefore cannot reproduce the infinite DC gain of an ideal $s^{-0.5}$ operator. The resulting residual steady-state error prolongs the final approach to the 2% settling band. This limitation is specific to the finite-band rational approximation and does not affect the phase-margin or transient-damping conclusions.

3.6 Approximation-Order and Local Approximation Checks

Because the fractional operators are implemented using finite rational approximations, the main conclusions must not depend on one arbitrary approximation order. Figure 9

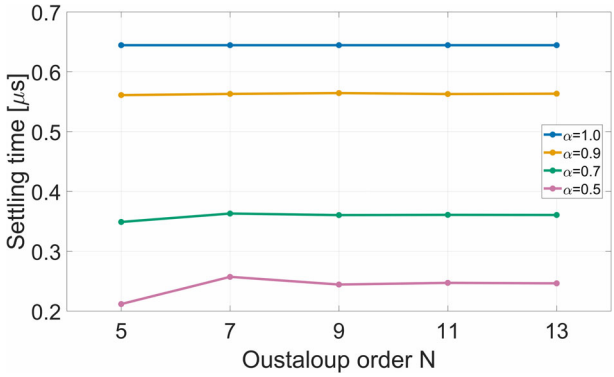


Fig. 9 Approximation-order sensitivity of equal-PM settling time. The main trends remain consistent from $N = 5$ to $N = 13$, with $N = 9$ used for the main manuscript results

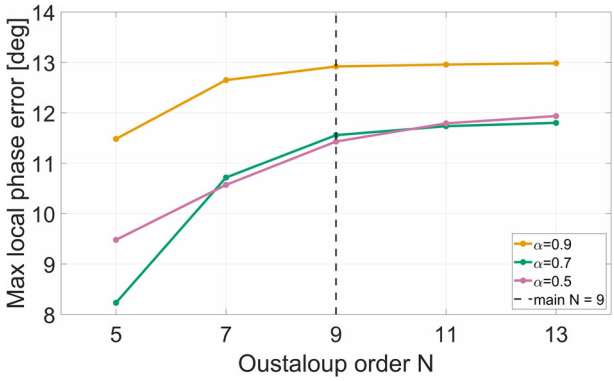


Fig. 10 Crossover-region Oustaloup phase error for the fractional operators. The vertical dashed line marks the main order $N = 9$. The maximum error over the plotted region is complemented in the text by the phase error evaluated exactly at each equal-PM gain-crossover frequency

shows the equal-PM settling-time sensitivity for $N = 5, 7, 9, 11, 13$. The ordering of the fractional responses remains stable across this range, and the $\alpha = 0.5$ settling-time spread over the tested orders is approximately 0.0454 μs.

Figure 10 shows the local phase error of the finite rational approximation around the relevant crossover region. For the main setting $N = 9$, the maximum local phase error over the plotted region is approximately 12.92°. The more directly relevant quantity for phase-margin assessment is the error exactly at each equal-PM gain crossover. Re-evaluation at those frequencies gives open-loop phase errors of $-0.614°$ at $\omega_{cp} = 2.5929 \times 10^7$ rad/s for $\alpha = 0.9$, $-0.655°$ at $\omega_{cp} = 3.0908 \times 10^7$ rad/s for $\alpha = 0.7$, and $-1.130°$ at $\omega_{cp} = 4.5381 \times 10^7$ rad/s for $\alpha = 0.5$. Thus, the crossover-specific error remains within 1.13° for all reported fractional cases, even though the worst-case error over the broader plotted region is larger. This distinction is important because PM is defined at the actual 0 dB crossover. The agreement across approximation orders and the LTspice validation below further support the stability of the reported conclusions.

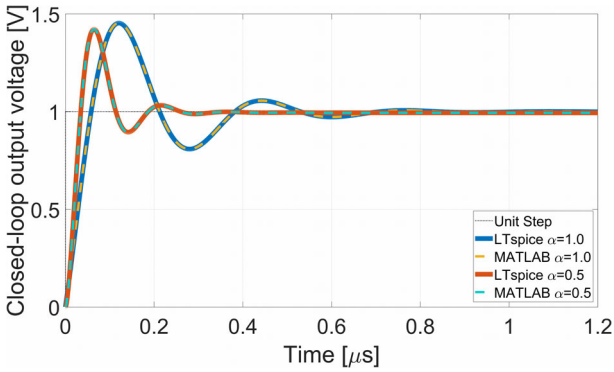


Fig. 11 LTspice behavioral-circuit validation against the MATLAB model. A 1 V reference step was applied, and the closed-loop plant output voltage was recorded. Solid lines denote LTspice simulations, while dashed lines denote MATLAB transfer-function simulations. The near overlap validates both the integer baseline and the $\alpha = 0.5$ fractional CPE-retuned controller at circuit-netlist level

3.7 LTspice Behavioral-Circuit Validation

Figure 11 compares the MATLAB transfer-function responses with LTspice behavioral-circuit simulations. The solid curves show LTspice results, and the dashed curves show MATLAB results. The close overlap confirms that the response is reproduced at circuit-netlist level using ideal AD-IC port equations and finite RC CPE emulation.

For $\alpha = 1.0$, LTspice produced a rise time of 45.400 ns, peak time of 120.500 ns, overshoot of 45.332%, and settling time of 0.64450 μs, compared with 45.405 ns, 120.410 ns, 45.332%, and 0.64446 μs in MATLAB. For $\alpha = 0.5$, the corresponding LTspice values were 24.500 ns, 64.630 ns, 43.186%, and 0.24443 μs, compared with 24.402 ns, 64.606 ns, 43.186%, and 0.24442 μs in MATLAB. These differences are negligible relative to the transient timescale and confirm that the fractional retuning result is not a MATLAB-only artifact.

The agreement in Fig. 11 validates the internal consistency of the modeling chain. The same controller law is obtained from the AD-IC port equations, the same plant transfer function is reproduced by the behavioral state-space plant, and the same finite fractional approximation is used in MATLAB and LTspice. This does not replace post-layout or fabricated-silicon verification, but it shows that the proposed fractional AD-IC circuit abstraction can be mapped into a circuit netlist without changing the reported closed-loop response. A complementary transistor-level frequency-response validation using the BSIM3 AD-IC subcircuit in PSpice is presented in Sect. 3.8.

3.8 PSpice Transistor-Level Frequency-Response Validation

To complement the behavioral LTspice validation in Sect. 3.7, a transistor-level frequency-response check was performed in PSpice using the TSMC 0.25 μm BSIM3 AD-IC subcircuit reported in the original integer-order study [24]. This additional step moves the validation one level closer to physical silicon by replacing the ideal

behavioral port equations with a twelve-transistor MOSFET implementation of the AD-IC operating from a ± 1.25 V supply.

The open-loop PID frequency response was obtained by applying a small-signal alternating-current (AC) source between the Y_1 and X_1 ports of the AD-IC and measuring the voltage at Y_1 across a 10 kΩ load resistor. The transistor topology, device dimensions, process model, supply levels, and bias conditions were taken from the original PSpice implementation in [24]. For the integer case ($\alpha = 1.0$), the two passive branches of the PID network used ordinary 80 pF capacitors. For the fractional case ($\alpha = 0.5$), both capacitors were replaced by the same ninth-order Oustaloup Foster RC ladder used in the MATLAB and LTspice analyses, with $Q_{\text{opt}} = 1.253 \times 10^6$ pF-equivalent.

Figure 12 shows the resulting gain and phase plots alongside the corresponding ideal transfer functions. The ideal curves are shown as reference transfer functions rather than as absolute-gain matches. Because the transistor-level test setup includes finite loading and source-impedance effects, the comparison emphasizes the characteristic frequency-response shape, gain-minimum location, and phase behavior rather than absolute magnitude coincidence.

For $\alpha = 1.0$ (panels a–b), the PSpice gain minimum occurs at 1.35 MHz, compared with 1.62 MHz for the ideal PID, a frequency ratio of 1.2 within the control band. The phase responses agree closely up to approximately 50 MHz, which is more than one decade above the gain-crossover frequency of 3.78 MHz. The divergence above 100 MHz is due to the finite transistor bandwidth of the TSMC 0.25 μm process and does not affect closed-loop performance, since the control-relevant frequency range is fully covered by the agreement region.

For $\alpha = 0.5$ (panels c–d), the results are shown up to 100 MHz, which lies within the Oustaloup approximation band $[10^3, 10^9]$ rad/s (approximately 159 Hz–159 MHz). Over the plotted control-relevant region, the transistor-level gain minimum and phase profile follow the ideal fractional $\text{PI}^{0.5}\text{D}^{0.5}$ transfer function. The gain crossover of the fractional loop lies at $\omega_{\text{cp}} \approx 4.54 \times 10^7$ rad/s (7.2 MHz), well within the agreement region. Above approximately 100 MHz in the present realization, the finite RC ladder begins to depart visibly from the target fractional response and the gain rolls off rather than continuing to rise. This behavior is consistent with the finite-band nature of rational CPE approximations [22] and is not a limitation of the ideal fractional AD-IC topology itself. In a physical implementation, the CPE approximation band would be chosen to cover the application-specific control bandwidth, which for the loop considered here is fully contained within the valid band. The high-frequency rolloff also provides a practical benefit by attenuating noise amplification from the derivative path at frequencies well above the control band.

Taken together, the transistor-level PSpice results provide supporting evidence that the characteristic AD-IC-based PID frequency-response trends are retained at the MOSFET circuit level within the frequency range relevant to control, for both the integer baseline and the fractional CPE-retuned design. This provides circuit-netlist evidence that is consistent with, and independent of, the behavioral LTspice validation in Sect. 3.7.

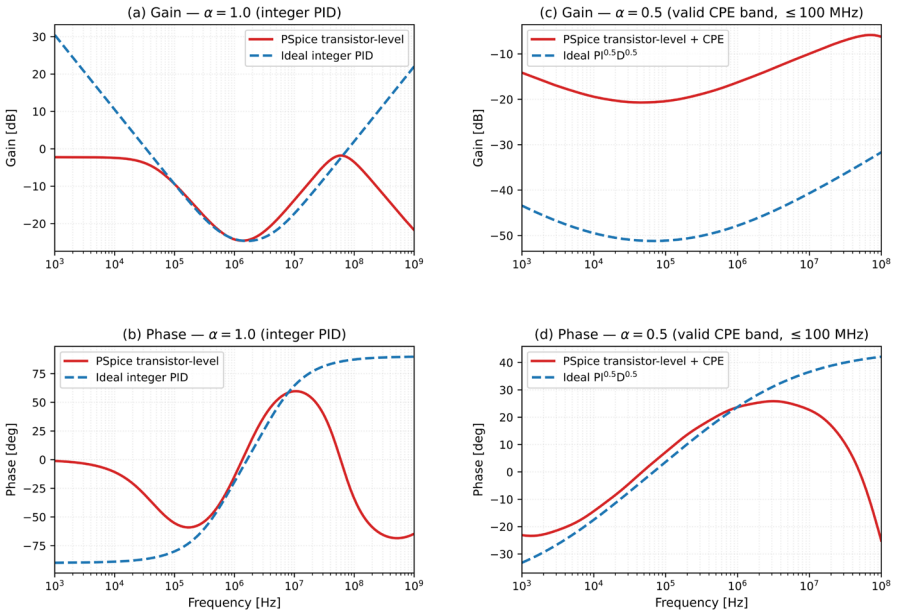


Fig. 12 PSpice transistor-level open-loop frequency-response validation of the AD-IC PID circuit using the TSMC 0.25 μm BSIM3 subcircuit. Panels (a)–(b): integer-order case ($\alpha = 1.0$, $C_1 = C_2 = 80$ pF) over the full sweep range 1 kHz–1 GHz. Panels (c)–(d): fractional case ($\alpha = 0.5$, $Q_{\text{opt}} = 1.253 \times 10^6$ pF-equivalent, $N = 9$ Oustaloup CPE ladder) shown up to 100 MHz, within the approximation band $[10^3, 10^9]$ rad/s (approximately 159 Hz–159 MHz). Solid curves: PSpice transistor-level simulation. Dashed curves: corresponding ideal integer or fractional $\text{PI}^\alpha\text{D}^\alpha$ transfer-function references. Because finite loading and source impedance introduce magnitude offsets in the transistor-level test setup, the comparison emphasizes frequency-response shape, gain-minimum location, and phase behavior rather than absolute gain coincidence. For $\alpha = 1.0$, the gain minimum locations agree within a frequency ratio of 1.2, and the phase responses match closely up to 50 MHz, well above the gain crossover at 3.78 MHz. For $\alpha = 0.5$, the transistor-level response follows the expected fractional trend over the control-relevant region, which fully covers the loop crossover at 7.2 MHz. The visible high-frequency deviations near the upper end of the plotted range reflect finite transistor bandwidth and finite-order RC approximation effects rather than the control-band behavior used for the reported stability conclusions

4 Discussion

4.1 Topology-Preserving Fractional Extension

The results support the view that the AD-IC voltage-mode PID topology admits a natural symmetric fractional extension. The important structural point is the preservation of $K_p = 1 + R_1/R_2$ after matched CPE substitution. In many analog PID topologies, replacing ordinary capacitors with fractional elements can alter several controller terms simultaneously and can make the physical interpretation of the coefficients less transparent. Here, the matched AD-IC structure keeps the proportional term tied to a resistor ratio, while the derivative-like and integral-like fractional terms are adjusted through one pseudo-capacitance scale. This is a useful property for ana-

log implementation because resistor ratios are usually easier to control and tune than absolute capacitance-like quantities.

The symmetric $PI^{\alpha}D^{\alpha}$ choice should therefore be understood as a topology-driven design, not merely as a reduction in parameter count. A fully general $PI^{\lambda}D^{\mu}$ controller would provide two independent orders, but it would require two different fractional elements in the two branches. That would increase implementation complexity, tolerance sensitivity, and the number of retuning variables. The present design deliberately preserves the matched-element philosophy of the reference AD-IC circuit and asks how far that structure can be extended before additional fractional degrees of freedom are needed.

4.2 CPE Retuning Law and Design Implications

The dense $Q_{\text{opt}}(\alpha)$ map in Fig. 3 is one of the practical outcomes of the study. The log-linear relationship does not imply that a literal capacitor grows to the reported pF-equivalent value when α decreases. Instead, it gives the pseudo-capacitance scale of the CPE model needed to restore the target PM after the phase of the fractional operator has changed. For $\alpha < 1$, Q carries order-dependent units and should be treated as a design parameter of the CPE emulator. In a passive ladder implementation, this target scale would be distributed across the ladder branches rather than realized as a single lumped capacitor.

The strong dependence of Q_{opt} on α also explains why direct substitution fails. If Q is kept fixed at the integer value, the derivative-like and integral-like paths no longer preserve the loop-shaping balance that produced the original PM. The resulting phase at crossover can move to an unstable region, as shown in Table 3. The proposed method should therefore be described as fractional retuning of the AD-IC topology, not simply fractional replacement of capacitors.

From a circuit-implementation perspective, the Q_{opt} scale is distributed across the branches of a passive RC ladder or active CPE emulator rather than realized as a single lumped element. A useful frequency-local interpretation is the effective capacitance

$$C_{\text{eff}}(\omega) = Q\omega^{\alpha-1}, \tag{26}$$

obtained by comparing $Y_{\text{CPE}} = Q(j\omega)^{\alpha}$ with an ordinary capacitive admittance at a selected frequency. For example, the apparently large $Q_{\text{opt}} = 1.253 \times 10^{6}$ pF-equivalent value for $\alpha = 0.5$ corresponds to $C_{\text{eff}} \approx 186$ pF at its nominal crossover $\omega_{\text{cp}} = 4.5381 \times 10^{7}$ rad/s. Thus, the pF-equivalent number should not be interpreted as a literal 1.253 μF on-chip capacitor. Passive ladders are conceptually straightforward and align with the RC approximation structure in Fig. 2, though they may require a wide component spread as the approximation band widens or α decreases significantly. Active CPE emulators based on operational transconductance amplifiers [39] can reduce this spread and allow electronic tuning of α, at the cost of additional active components and their associated bandwidth and noise constraints. Recent work has also demonstrated electronically tunable fractional-order elements based on distributed metal-oxide-semiconductor (MOS) structures, showing that integrated

CPE-like behavior need not be restricted to large discrete RC ladders [18]. Foster-network SPICE synthesis provides another practical route for constructing finite-band CPE behavior from ordinary RC elements [44]. The preferred implementation route will depend on the target frequency range, available technology, and the degree of tunability required.

4.3 Equal-Crossover Transient Behavior

The equal-PM step response in Fig. 4 shows a large nominal settling-time reduction, but part of that improvement comes from the increased gain-crossover frequency. This is expected because lower α values shift the equal-PM designs to higher crossover frequencies. The equal-PM and equal-ω_{cp} result in Fig. 5 is therefore the more conservative evidence for a true phase-shaping contribution. In that comparison, the initial rise does not improve for $\alpha = 0.7$ and $\alpha = 0.5$, yet the settling time remains shorter than the integer baseline.

This behavior is consistent with the frequency-domain response in Fig. 6. The fractional controller does not simply make the loop faster. It changes the local phase profile around crossover and modifies the damping of the later oscillatory tail. The equal-crossover response is therefore best interpreted as a reduction in post-peak oscillatory persistence, not as a shorter initial rise time. This distinction is important because it prevents overclaiming and aligns the time-domain evidence with the iso-damping interpretation.

4.4 Plant-Gain Robustness and Uncertainty

The plant-gain sweep in Fig. 7 is the clearest robustness result. A pure four-point scatter of PM standard deviation would not have added much information, so the full gain-sweep curves were retained instead. These curves show how PM changes continuously as the plant gain is perturbed. The integer design has a steep PM change and loses stability for sufficiently reduced gain. The lower-order fractional designs, especially $\alpha = 0.5$, keep the phase margin much closer to the target value over the tested range.

The time-domain evidence in Fig. 8 and Table 7 provides a direct visual complement to the PM sweep. At $K = 0.25\times$, the integer controller produces sustained oscillations that persist for approximately 7.8 μs, while the fractional $\alpha = 0.5$ controller yields a bounded, decaying response with overshoot (39.1%) close to its nominal value (41.1%). The PM variation across the three tested gain conditions is 61.1° for the integer design and only 2.6° for the fractional design. At the nominal operating point, the fractional controller also achieves a settling time approximately three times shorter than the integer design (0.206 μs versus 0.644 μs), consistent with the equal-PM results in Table 4. This combination of frequency-domain and time-domain evidence supports the iso-damping interpretation at both levels of analysis.

The Monte Carlo results in Table 6 should be interpreted carefully. The 100% stability rate is encouraging, but it is not by itself the main robustness claim because the tested uncertainty bounds are finite. The more meaningful trend is the narrowing

of the PM percentile interval as α decreases. This is consistent with the deterministic gain-sweep result and supports the iso-damping interpretation. The Monte Carlo result therefore acts as a perturbation check around the deterministic mechanism rather than as a proof of unconditional robustness. The retuning law in Eq. (23) was derived for the specific plant in Eq. (13), so the $Q_{\mathrm{opt}}(\alpha)$ values and PM trends reported here should be interpreted within that scope; whether the log-linear map generalizes to substantially different plant families is a natural direction for future work.

4.5 Role of Finite Approximation Checks

The approximation-order and crossover-region error results are retained in the main text because they define the numerical status of the fractional simulations. Fractional-order control papers can be weakened if the operator approximation is treated as a hidden numerical detail. Here, the approximation order is varied explicitly in Fig. 9, and the crossover-region phase error is quantified in Fig. 10. The use of $N = 9$ is therefore supported by an explicit order sweep rather than selected in isolation. The settling trends are not identical at every order, especially for $\alpha = 0.5$, but the ordering of the designs and the main conclusion remain stable across the tested range; at the actual equal-PM crossover frequencies, the $N = 9$ open-loop phase error remains no larger than $1.13°$.

The local phase error reported for $N = 9$ is not zero, so the finite rational realization should not be interpreted as an ideal fractional operator implemented over an infinite frequency range. The appropriate claim is narrower and more defensible: within the selected finite-band Oustaloup approximation and its RC-emulator equivalent, the fractional AD-IC retuning produces the reported transient and robustness behavior. This claim is supported by both approximation-order sensitivity and LTspice behavioral-circuit validation.

4.6 Circuit-Netlist Validation Scope

The LTspice validation in Fig. 11 materially strengthens the manuscript because it shows that the result is not only a MATLAB transfer-function artifact. The AD-IC equations were implemented with ideal behavioral voltage sources, and the fractional element was implemented with a finite RC CPE emulator derived from the same ninth-order approximation used in MATLAB. The nearly identical MATLAB and LTspice waveforms demonstrate consistency between the transfer-function model and a circuit-netlist realization of the same idealized topology.

The PSpice transistor-level validation in Fig. 12 extends this evidence to the MOSFET circuit level. Within the control-relevant frequency band, the twelve-transistor BSIM3 AD-IC implementation shows frequency-response behavior consistent with the expected PID gain and phase trends for both $\alpha = 1.0$ and $\alpha = 0.5$. This two-level validation strategy combines behavioral netlist consistency in LTspice with transistor-level frequency-response agreement in PSpice. It provides a coherent chain of evidence from the analytical model to the physical circuit while honestly acknowledging the boundaries of each validation step.

This validation should not be overstated. It does not establish post-layout robustness, fabricated-silicon performance, device-noise tolerance, or mismatch tolerance in the RC ladder. The behavioral LTspice model also omits active-device nonidealities by construction, while the transistor-level PSpice check is limited to open-loop frequency-response shape rather than a full transistor-level closed-loop transient validation. The evidence therefore supports a narrower conclusion: the analytical controller law maps consistently to the behavioral circuit model, and the transistor-level AD-IC subcircuit exhibits the relevant frequency-response trend over the control band. For a circuits-oriented manuscript, this provides a useful bridge between transfer-function analysis and later post-layout or experimental validation.

5 Conclusion

A fractional-order extension of a single-AD-IC voltage-mode PID controller was developed and evaluated. The integer baseline reproduced the reference controlled-plant response with a maximum error below 0.26%. Direct fractional substitution with fixed Q destabilized the tested fractional cases, showing that CPE replacement alone is not a valid design method. A one-parameter CPE pseudo-capacitance retuning strategy was then introduced to recover the reference PM for each fractional order. With a ninth-order Oustaloup approximation, the retuned fractional controllers reduced settling time by up to 62.07% under equal PM and by 27.37% under equal PM and equal gain crossover. Plant-gain sweep analysis showed a 93.40% reduction in PM standard deviation for $\alpha = 0.5$, and all 300 Monte Carlo trials per reported order remained stable under the tested uncertainty ranges. Time-domain step responses under a fourfold plant gain reduction demonstrated that the fractional $\alpha = 0.5$ controller maintains bounded, well-damped behavior (PM = 37.4°, overshoot $\approx$ 39%) while the integer controller produces sustained oscillations (PM = 5.7°), confirming the iso-damping property at the time-domain level. Behavioral LTspice validation reproduced the MATLAB responses for both $\alpha = 1.0$ and $\alpha = 0.5$, and a complementary PSpice transistor-level frequency-response check provided circuit-level evidence supporting the expected PID gain and phase trends within the control-relevant frequency band for both designs. Overall, CPE retuning adds useful phase-shaping and iso-damping capability to the compact AD-IC PID architecture. The design route established here consists of matched CPE substitution, one-parameter Q_{opt} retuning, equal-crossover fairness verification, and two-level circuit-netlist validation. This reproducible framework provides a basis for future work including full transistor-level closed-loop simulation, tolerance-aware CPE synthesis, and experimental validation of the fractional topology.

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Data Availability The generated tables and figure data supporting the findings of this study are available from the corresponding author upon reasonable request.

Code Availability The MATLAB scripts and the LTspice and PSpice netlists used in this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethics Approval and Consent to Participate Not applicable. This study did not involve human participants or animals.

Consent for Publication Both authors approved the final manuscript and consent to its publication.

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References

1. K.H. Ang, G. Chong, Y. Li, PID control system analysis, design, and technology. IEEE Trans. Control Syst. Technol. **13**(4), 559–576 (2005). https://doi.org/10.1109/TCST.2005.847331
2. K.J. Åström, T. Hägglund, *Advanced PID Control* (ISA, Research Triangle Park, NC, USA, 2006)
3. U.E. Ayten, E. Yüce, S. Minaei, A voltage-mode PID controller using a single CFOA and only grounded capacitors. Microelectron. J. **81**, 84–93 (2018). https://doi.org/10.1016/j.mejo.2018.09.010
4. T.K. Bashishtha, V.P. Singh, U.K. Yadav, U.K. Sahu, Fractional-order PID controllers and applications: a comprehensive survey. Annu. Rev. Control. **60**, 101013 (2025). https://doi.org/10.1016/j.arcontrol.2025.101013
5. H.W. Bode, *Network Analysis and Feedback Amplifier Design* (Van Nostrand, New York, NY, USA, 1945)
6. Z.G. Çam Taşkıran, H. Sedef, F. Anday, A new PID controller circuit design using CFOAs. Circuits Syst. Signal Process. **40**(3), 1166–1182 (2021). https://doi.org/10.1007/s00034-020-01540-5
7. G.E. Carlson, C.A. Halijak, Approximation of fractional capacitors $(1/s)^{1/n}$ by a regular Newton process. IEEE Trans. Circuit Theory **11**(2), 210–213 (1964). https://doi.org/10.1109/TCT.1964.1082270
8. A. Charef, H.H. Sun, Y.Y. Tsao, B. Onaral, Fractal system as represented by singularity function. IEEE Trans. Autom. Control **37**(9), 1465–1470 (1992). https://doi.org/10.1109/9.159595
9. Y.Q. Chen, T. Bhaskaran, D. Xue, Practical tuning rule development for fractional order proportional and integral controllers. J. Comput. Nonlinear Dyn. **3**(2), 021403 (2008). https://doi.org/10.1115/1.2833934
10. S.C. Dutta Roy, On the realization of a constant argument immittance or a fractional operator. IEEE Trans. Circuit Theory **14**(3), 264–274 (1967). https://doi.org/10.1109/TCT.1967.1082706
11. R. El-Khazali, On the biquadratic approximation of fractional-order Laplacian operators. Analog Integr. Circ. Sig. Process **82**(3), 503–517 (2015). https://doi.org/10.1007/s10470-014-0432-8
12. A.S. Elwakil, Fractional-order circuits and systems: an emerging interdisciplinary research area. IEEE Circuits Syst. Mag. **10**(4), 40–50 (2010). https://doi.org/10.1109/MCAS.2010.938637

13. C. Erdal, A. Toker, C. Acar, OTA-C based proportional-integral-derivative (PID) controller and calculating optimum parameter tolerances. Turk. J. Electr. Eng. Comput. Sci. **9**(2), 189–198 (2001)
14. T.J. Freeborn, A survey of fractional-order circuit models for biology and biomedicine. IEEE J. Emerg. Sel. Top. Circuits Syst. **3**(3), 416–424 (2013). https://doi.org/10.1109/JETCAS.2013.2265797
15. I.C. Göknar, E. Minayi, Realizations of mutative 4-ports and their applications to memstor simulations. Analog Integr. Circ. Sig. Process **81**(1), 29–42 (2014). https://doi.org/10.1007/s10470-014-0354-5
16. I.C. Göknar, E. Minayi, A Rectifier Circuit Using Add-Differentiate IC with a Minimal Number of CMOS Transistors, in *2018 25th IEEE International Conference on Electronics, Circuits and Systems (ICECS)*, Bordeaux, France, 2018, pp. 257–260, https://doi.org/10.1109/ICECS.2018.8617914
17. A. Kartci, N. Herencsar, J.A.T. Machado, L. Brancik, History and progress of fractional-order element passive emulators: a review. Radioengineering **29**(2), 296–304 (2020). https://doi.org/10.13164/re.2020.0296
18. D. Kubanek, A. Shadrin, J. Dvorak, J. Jerabek, P. Seda, R. Sotner, V. Kledrowetz, C. Christie, T.J. Freeborn, P.A. Ushakov, Design of electronically tunable fractional-order elements based on distributed MOS transistor structures. Sci. Rep. **16**, 19122 (2026). https://doi.org/10.1038/s41598-026-48269-w
19. B.C. Kuo, *Automatic Control Systems*, 7th edn. (Prentice Hall, Englewood Cliffs, NJ, USA, 1995)
20. P. Lanusse, A. Oustaloup, V. Pommier-Budinger, Stability of closed loop fractional order systems and definition of damping contours for the design of controllers. Int. J. Bifurc Chaos **22**(4), 1230013 (2012). https://doi.org/10.1142/S0218127412300133
21. G. Maione, P. Lino, New tuning rules for fractional PI$^{\alpha}$ controllers. Nonlinear Dyn. **49**(1–2), 251–257 (2007). https://doi.org/10.1007/s11071-006-9125-x
22. F. Merrikh-Bayat, Rules for selecting the parameters of Oustaloup recursive approximation for the simulation of linear feedback systems containing PI$^{\lambda}$D$^{\mu}$ controller. Commun. Nonlinear Sci. Numer. Simul. **17**(4), 1852–1861 (2012). https://doi.org/10.1016/j.cnsns.2011.08.042
23. S. Minaei, I.C. Göknar, M. Yıldız, E. Yüce, Memstor, memstance simulations via a versatile 4-port built with new adder and subtractor circuits. Int. J. Electron. **102**(6), 911–931 (2015). https://doi.org/10.1080/00207217.2014.942890
24. E. Minayi, I.C. Göknar, Design of voltage-mode and current-mode PID controllers using a single Add-Differentiate IC (AD-IC). Circuits Syst. Signal Process. **45**(7), 5119–5139 (2026). https://doi.org/10.1007/s00034-025-03456-4
25. C.A. Monje, Y.Q. Chen, B.M. Vinagre, D. Xue, V. Feliu, Fractional-Order Systems and Controls: Fundamentals and Applications (Springer, 2010), https://doi.org/10.1007/978-1-84996-335-0
26. K. Ogata, *Modern Control Engineering*, 5th edn. (Prentice Hall, Upper Saddle River, NJ, USA, 2010)
27. K.B. Oldham, J. Spanier, *The Fractional Calculus: Theory and Applications of Differentiation and Integration to Arbitrary Order* (Academic Press, New York, NY, USA, 1974)
28. U.C. Oruçoğlu, E. Özer, F. Kaçar, VDIBA-based current-mode PID controller design. J. Circuits Syst. Comput. **32**(17), 2350288 (2023). https://doi.org/10.1142/S0218126623502882
29. A. Oustaloup, *La Commande CRONE: Commande Robuste d'Ordre Non Entier* (Hermès, Paris, France, 1991)
30. E. Özer, F. Kaçar, Design of voltage-mode PID controller using a single voltage differencing current conveyor (VDCC). Analog Integr. Circ. Sig. Process **109**(1), 11–27 (2021). https://doi.org/10.1007/s10470-021-01880-5
31. E. Özer, A. Sayın, F. Kaçar, Voltage-mode PID controller design employing canonical number of active and passive elements. Analog Integr. Circ. Sig. Process **113**(3), 361–371 (2022). https://doi.org/10.1007/s10470-022-02057-4
32. F. Padula, A. Visioli, Optimal tuning rules for proportional-integral-derivative and fractional-order proportional-integral-derivative controllers for integral and unstable processes. IET Control Theory Appl. **6**(6), 776–786 (2012). https://doi.org/10.1049/iet-cta.2011.0419
33. I. Petráš, Fractional-Order Nonlinear Systems: Modeling, Analysis and Simulation (Springer, 2011), https://doi.org/10.1007/978-3-642-18101-6
34. I. Podlubny, Fractional-order systems and PI$^{\lambda}$D$^{\mu}$ controllers. IEEE Trans. Autom. Control **44**(1), 208–214 (1999). https://doi.org/10.1109/9.739144
35. P. Prommee, K. Angkeaw, High performance electronically tunable log-domain current-mode PID controller. Microelectron. J. **72**, 126–137 (2018). https://doi.org/10.1016/j.mejo.2017.09.008
36. N. Roongmuanpha, J. Satansup, T. Pukkalanun, W. Tangsrirat, Design of mixed-mode analog PID controller with CFOAs. Sensors **24**(10), 3125 (2024). https://doi.org/10.3390/s24103125

37. J. Sabatier, P. Lanusse, P. Melchior, A. Oustaloup, Fractional Order Differentiation and Robust Control Design: CRONE, H-Infinity and Motion Control (Springer, 2015), https://doi.org/10.1007/978-94-017-9807-5
38. W. Tangsrirat, Voltage-mode analog PID controller using a single z-copy current follower transconductance amplifier (ZC-CFTA). Informacije MIDEM **45**(3), 175–179 (2015)
39. G. Tsirimokou, C. Psychalinos, A.S. Elwakil, Emulation of a constant phase element using operational transconductance amplifiers. Analog Integr. Circ. Sig. Process **85**(3), 413–423 (2015). https://doi.org/10.1007/s10470-015-0626-8
40. D. Valério, J.S. da Costa, Tuning of fractional PID controllers with Ziegler-Nichols-type rules. Signal Process. **86**(10), 2771–2784 (2006). https://doi.org/10.1016/j.sigpro.2006.02.020
41. B.M. Vinagre, I. Podlubny, L. Dorčák, V. Feliu, On fractional PID controllers: a frequency domain approach. IFAC Proc. Vol. **33**(4), 51–56 (2000). https://doi.org/10.1016/S1474-6670(17)38220-4
42. A. Visioli, *Practical PID Control* (Springer, London, UK, 2006). https://doi.org/10.1007/1-84628-586-0
43. S. Westerlund, L. Ekstam, Capacitor theory. IEEE Trans. Dielectr. Electr. Insul. **1**(5), 826–839 (1994). https://doi.org/10.1109/94.326654
44. M. Wilson, L. Cowie, V. Farrow, M. Cree, J. Scott, Rapid time-domain simulation of fractional capacitors with SPICE. J. Comput. Electron. **23**, 677–689 (2024). https://doi.org/10.1007/s10825-024-02160-x
45. A.R. Ynineb, M.D. Mihai, E. Yumuk, H. Farbakhsh, G. Ben Othman, R. De Keyser, C. Muresan, I. Birs, D. Copot, C.M. Ionescu, Fractional order PID control of propofol dosage and optimization in lean and obese patients. IFAC-PapersOnLine **58**(7), 228–233 (2024). https://doi.org/10.1016/j.ifacol.2024.08.039
46. A. Yonezawa, H. Yonezawa, S. Yahagi, I. Kajiwara, Practical one-shot data-driven design of fractional-order PID controller: fictitious reference signal approach. ISA Trans. **152**, 208–216 (2024). https://doi.org/10.1016/j.isatra.2024.07.001
47. E. Yüce, S. Tokat, A. Kızılkaya, O. Çiçekoğlu, CCII-based PID controllers employing grounded passive components. AEU-Int. J. Electron. C. **60**(5), 399–403 (2006). https://doi.org/10.1016/j.aeue.2005.03.017
48. E. Yüce, S. Minaei, New CCII-based versatile structure for realizing PID controller and instrumentation amplifier. Microelectron. J. **41**(5), 311–316 (2010). https://doi.org/10.1016/j.mejo.2010.03.008
49. J.G. Ziegler, N.B. Nichols, Optimum settings for automatic controllers. Trans. ASME **64**(8), 759–765 (1942). https://doi.org/10.1115/1.4019264

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